Can Recommending Real-World Interrogation Techniques Help LLMs Moderate Extremist Views?

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ABSTRACT

Moderating extremist or entrenched views online remains a persistent challenge. We ask whether a retrieval-augmented generation (RAG) recommender that selects real-world interrogation and debate techniques at each conversational turn can produce more reliably *directed* stance change in a simulated LLM suspect than an unguided baseline. We build a three-role system (Suspect, Interrogator, and Judge) and evaluate it with a quad experimental design (matched control and treatment pairs) across four ballot-measure topics, including one drawn from verbatim arguments of the Davis Measure V June 2026 special election. Across 80 experimental legs, the treatment interrogator achieves **75% directional accuracy** versus **35%** for the unguided control, a 40-percentage-point improvement ($p < 0.001$). Pooled stance-change *magnitude*, by contrast, is nearly identical across conditions: the effect is on the direction and smoothness of movement, not its size. The largest single-topic gap is Transit (10% → 90%). Measure V, our real-data condition, replicates the gain (30% → 70%), confirming that technique recommendation transfers to genuine public-debate arguments. These results suggest that RAG-guided technique recommendation is a viable component for LLM-based persuasion and moderation systems.

KEYWORDS

retrieval-augmented generation, recommender systems, stance change, LLM agents, content moderation, persuasion

1 INTRODUCTION

The proliferation of online extremism and entrenched misinformation has motivated sustained research into automated moderation [3, 6]. Most deployed systems operate at the content level (flagging, removing, or down-ranking objectionable posts) but do not attempt to change the underlying beliefs that motivate harmful speech. A growing body of work instead explores *conversational intervention*: engaging an individual in dialogue to shift their stated position [1]. Early results are promising but rely on general-purpose language models with no principled mechanism for selecting which argumentative strategy to deploy next. This work is situated within a broader research vision of AI as a facilitator of constructive dialogue and healthy discourse, one that addresses polarized views not by removing speech after the fact, but by engaging with the underlying beliefs that motivate it.

Recommender systems offer a natural framing for this problem. The *query* is the suspect's current conversational state (most recent utterance, inferred values, running argument history); the *item catalog* is a curated corpus of interrogation and debate techniques drawn from professional practice; and the *recommendation* drives the interrogator's next move. This framing connects to the broader RAG literature, where retrieved context improves generation quality [5], and to multi-agent debate research, where structured disagreement improves LLM reasoning [2].

We make the following contributions:

- A three-role multi-agent architecture (Suspect, Interrogator, Judge) with a two-channel RAG recommender: one channel for technique selection and one for topic-aligned argument retrieval.
- A *quad experimental design* that pairs direction-matched control and treatment legs, enabling within-quad comparison that mitigates judge scaling bias.
- An evaluation across four ballot-measure topics including one using verbatim real-world arguments (Davis Measure V, June 2026), demonstrating that technique recommendation generalizes beyond synthetic data.
- Empirical evidence that the two-channel RAG recommender (technique *and* argument retrieval together) improves directional accuracy from 35% to 75% across 80 experimental legs ($p < 0.001$, pooled). We are explicit that the design does not isolate the technique channel from the argument channel; an arguments-only ablation is the most important follow-up experiment.

2 RELATED WORK

2.1 Conversational Persuasion and Stance Change

Costello et al. [1] show that a single AI-driven conversation can durably reduce conspiracy beliefs, with effects persisting two months later. Their approach uses a general-purpose LLM without a curated technique corpus. Our work extends this by asking whether a *recommender* that selects among professionally documented techniques can improve on the unguided baseline. Where Costello et al. treat persuasion strategy as implicit in the model's pretraining, we make strategy selection an explicit recommendation problem, separating the choice of technique from its execution so that each can be controlled and studied independently.

2.2 LLM-Based Content Moderation

Kumar et al. [6] demonstrate that instruction-tuned LLMs outperform traditional classifiers on toxicity detection but remain sensitive to context shifts. Hong et al. [4] show through curiosity-driven red-teaming that LLMs can be induced to produce unsafe outputs under adversarial prompts, motivating the need for robust intervention mechanisms beyond passive filtering. Our work complements this detection-oriented line by operating at a different layer: rather than flagging content after it appears, we attempt to shift the underlying beliefs that produce it, a proactive complement to reactive content moderation. Adaptively recommending techniques at each turn may also contribute a form of robustness: a structured technique corpus constrains the interrogator's behaviour, reducing the risk of an LLM spiralling into incoherent or off-target responses when the conversation takes an unexpected turn.

2.3 Multi-Agent Debate

Estornell and Liu [2] study multi-agent debate as a mechanism for improving LLM reasoning quality. They find that when participating models share similar capabilities or come from the same training distribution, debates can stall: rather than reaching better answers through disagreement, agents converge on the majority opinion embedded in their shared training data, including shared misconceptions. This means that adding more debating agents does not guarantee correctness when all agents share the same blind spots. Unlike Estornell and Liu's multi-agent reasoning setup, our system uses structured dialogue for belief intervention rather than answer verification. However, our use of a single model family (llama3.1:8b) for all three roles introduces the same shared-bias concern; we treat this as a known limitation.

2.4 Retrieval-Augmented Generation

Izacard et al. [5] establish that retrieved passages substantially improve generation on knowledge-intensive tasks; this grounding motivates both retrieval channels in our system. We extend standard RAG in one respect: retrieval is *turn-aware*. Rather than querying a fixed index, our phase-filtered technique recommender (Section 3.5) changes which techniques are eligible as the conversation advances, a constraint we draw from interrogation practice rather than from the retrieval literature.

2.5 Network Science and Influence Propagation

Ribeiro et al. [9] characterize hateful user communities on Twitter and find that such users are not isolated actors: they form tightly connected communities with high centrality in retweet networks, with a small number of high-centrality accounts responsible for a disproportionate share of content propagation. Yu et al. [11] and Sun et al. [10] provide dynamic graph-learning tools that model how influence propagates through such networks over time. Together, these findings suggest a natural upstream component for systems like ours: using network analysis to identify high-centrality individuals before initiating dialogue, rather than waiting for organic encounter. Our current evaluation does not include this targeting stage; we return to it in the Conclusion.

3 SYSTEM DESIGN

3.1 Three-Role Architecture

The system comprises three LLM roles, each running as a stateful agent against a local Ollama instance (llama3.1:8b throughout, 180 s per call timeout):

Suspect. Initialized at one of the two stance *poles*, chosen at random ($d_0 \in \{-2, +2\}$: strongly against or strongly for), together with a latent persona (values, communication style, and position on the topic), and a bank of aligned arguments drawn from the topic card (a structured YAML file that encodes the debate question, arguments for both sides, and suspect values; see Section 3.3). The Suspect responds in character and updates its internal model as the dialogue progresses. Only arguments consistent with the current stance are surfaced (`aligned_only` mode), preventing incoherent position reversals mid-utterance.

Interrogator. Operates in one of two conditions:

- *Control:* Receives only the topic question and Suspect utterance; no technique guidance.
- *Treatment:* Runs a two-stage RAG pipeline (Section 3.4) that retrieves a technique recommendation and a topic-aligned counter-argument before generating each response.

Because the control receives *neither* retrieval channel, this contrast measures the *combined* effect of technique guidance and grounded argument retrieval. It does not, on its own, isolate the contribution of technique selection from that of argument retrieval. We are explicit about this throughout and identify the missing arguments-only condition as the most important follow-up experiment (Section 6.4).

Judge. Scores each Suspect utterance on a five-point scale from -2 (strong disagreement) to $+2$ (strong agreement) against the topic proposition. Scoring is batched across all legs of a quad in a single call to ensure a consistent scale; out-of-range outputs are clamped and a per-utterance fallback is used when batch decoding fails (see Section 3.6).

3.2 Technique Corpus

The technique corpus consists of 18 Markdown cards drawn from three professional frameworks: the Reid Technique (5 cards covering confrontation, minimization, and theme development), the Scharff Elicitation Technique (7 cards), and structured academic debate (6 cards covering steelmanning, objection handling, and Socratic questioning). Each card is structured with YAML front-matter followed by a prose body. An example front-matter is shown below:

```
---
name: Rapport Building
phase: early
triggers:
  - beginning of a conversation before position
    has been established
  - suspect appears suspicious of motives
pairs_with: [orbit, oars, motivational_interviewing]
in_tension_with: [surfacing_contradictions,
```

```
                  scharff_technique]
out_of_bounds:
  - Do not manufacture false common ground
  - Do not skip rapport-building in favour of
    immediate persuasion — it consistently backfires
---
```

The phase field constrains when a card is eligible for retrieval; `pairs_with` and `in_tension_with` encode compatibility metadata that the Stage-1 selector uses to choose a complementary technique; `out_of_bounds` constraints are passed verbatim to the Stage-2 generator as hard guardrails.

3.3 Two-Channel RAG Recommender

Infrastructure. All 18 technique card bodies and all topic arguments are stored in ChromaDB, a persistent vector database that supports approximate nearest-neighbor search over dense embeddings. At each conversational turn, ChromaDB returns the k most semantically similar documents to the current query; retrieval latency is negligible relative to the cost of LLM generation.

Documents are embedded with `all-MiniLM-L6-v2`, a sentence-transformer model chosen for two practical reasons: (1) it runs entirely on CPU with no network dependency, matching our offline Ollama setup, and (2) its 384-dimensional embeddings are well-suited to short, precise prose, matching the format of both technique card bodies (50–400 tokens) and individual debate arguments. A domain-adapted embedding model trained on interrogation or debate corpora would likely improve retrieval precision, particularly for nuanced state distinctions such as "suspect is deflecting" versus "suspect is emotionally engaged"; we treat this as a direction for future work.

Channel 1: Technique Recommender. The bodies of the 18 technique cards are embedded individually and stored in a dedicated collection. At each turn, the query is the concatenation of the Suspect's latest utterance with the most recent conversation context. After phase filtering (Section 3.5), the top-k=5 semantically matching technique cards are retrieved. The value k=5 was chosen to give the Stage-1 selector meaningful choice across the available phase-eligible cards (typically 6–10 after filtering) without diluting the relevance signal. Passing fewer candidates (k=2) risks missing a well-matched technique; more (k=8) risks including cards whose trigger conditions do not match the current conversational state.

Channel 2: Argument Recommender. Each topic card contributes argument strings for both sides, embedded individually. The top-k=3 arguments aligned with the Interrogator's stance are retrieved per turn. k=3 keeps the Stage-2 generator focused: more arguments risk introducing conflicting rhetorical frames, while fewer risk missing the most contextually relevant counter-argument. Both k values were selected heuristically given our corpus scale and were not swept; sensitivity analysis is left for future work.

3.4 Treatment Pipeline: Two-Stage Recommendation

The treatment Interrogator separates *strategy selection* from *response generation* into two sequential LLM calls. This separation is a deliberate design choice: selection requires analytical reasoning (classify the suspect's orientation, rank techniques, decide persona disclosure), while generation requires fluent, naturalistic output. Combining both into one call risks the generator drifting from the selected technique under the pressure of producing fluent text. The two-stage structure also makes the system modular: swapping the topic card changes the argument bank without affecting the selector logic, and the selector can be retargeted to a new technique corpus without touching the generator prompt.

Algorithms 1 and 2 contrast the two conditions:

Algorithm 1 Control Interrogator (per turn)

Require: topic question Q, conversation history H, suspect utterance u
Ensure: interrogator response r
Append u to H
$r \leftarrow \text{LLM}(\text{sys} = \text{ControlPrompt}(Q),\ H)$ ▷ sys: system prompt built from Q
Append r to H
return r

Algorithm 2 Treatment Interrogator (per turn)

Require: topic card C, history H, suspect utterance u, turn t, budget T, suspect model $\mathcal{M}$, persona state $\mathcal{P}$
Ensure: interrogator response r
phase $\leftarrow$ PhaseOf(t, T) ▷ opening / middle / closing
$\mathbf{T}_{\text{top}} \leftarrow \text{ChromaDB.query}(u \oplus H,\ k{=}5,\ \text{filter=phase})$ ▷ Channel 1; $u \oplus H$: suspect utterance concatenated with recent history
$\mathbf{A}_{\text{top}} \leftarrow \text{ChromaDB.query}(u,\ k{=}3,\ \text{side=int\_stance})$ ▷ Channel 2
Append u to H
Stage 1 (Selector):
$S \leftarrow \text{LLM}(\text{SelectorPrompt},\ H,\ \mathcal{M},\ \mathcal{P},\ \mathbf{T}_{\text{top}},\ \mathbf{A}_{\text{top}})$ ▷ JSON: technique, orientation, persona reveal
Update $\mathcal{M}$ with orientation, inferred values, new arguments from S
Update $\mathcal{P}$: move disclosed detail from latent to revealed
Stage 2 (Generator):
$r \leftarrow \text{LLM}(\text{GeneratorPrompt},\ H,\ S.\text{technique},\ \mathbf{A}_{\text{top}},\ \mathcal{P}.\text{revealed})$
Append r to H
return r

Two-channel combination in Stage 2. Both channels' outputs are passed simultaneously to the Stage-2 generator: the full prose body and out-of-bounds constraints of the selected technique card (Channel 1), and the top-3 argument strings as an "available to deploy" list (Channel 2). The generator has discretion over argument usage: it may rephrase a retrieved argument positively, ignore it if it would create tension with the selected technique, or surface it indirectly using the *Scharff framing* described below. For example, if Channel 1 selects *Rapport Building* (which discourages direct challenge) and Channel 2 retrieves a sharp factual rebuttal, the generator typically softens the rebuttal into a curiosity-driven observation.

Scharff-style indirect argument deployment. The Stage-2 generator is instructed to deploy retrieved arguments indirectly rather than citing them as facts. This reflects the Scharff Technique, developed by WWII German intelligence officer Hanns Scharff and later formalized by Oleszkiewicz, Granhag, and Kleinman [8]: instead of asking direct questions (which reveal what the interrogator does not know) or stating facts (which trigger counter-argument), the interrogator implies familiarity and invites the suspect to fill in gaps. In our setting this means framing retrieved arguments as things the Interrogator has "noticed" or "heard others say" (*"I've seen that the affordability numbers on similar projects tend to shift once you factor in the subsidized units…"*), rather than as direct rebuttals. This keeps the conversation feeling like collaborative exploration rather than persuasion, which reduces the Suspect's defensive responses.

3.5 Phase-Filtered Retrieval

Technique retrieval is phase-aware: the current turn number is mapped to a conversational phase (`opening`: turns 1–2, `middle`: 3–4, `closing`: 5–6), and technique cards whose `phase` annotation does not match are excluded before the semantic search. This design reflects established practice in professional interrogation: rapport-building is most effective before substantive disagreement is introduced; challenge techniques (e.g., Surfacing Contradictions, Cognitive Dissonance) require trust to have been established and are counterproductive in opening turns; closing techniques focus on consolidation rather than new information. Enforcing phase-appropriate retrieval also keeps evaluations to a practical 6-turn length, consistent with real online conversations, which rarely sustain a single debate thread much longer. We do not report a controlled comparison between phased and unphased retrieval (i.e., an experiment that disables the phase filter while holding all else constant); this is acknowledged as a limitation.

3.6 Robustness Fixes

During evaluation on the Measure V topic, whose proposition is substantially longer than the synthetic topics, the Judge LLM occasionally produced out-of-range scores (e.g., 8 on a $[-2, 2]$ scale) or returned arrays of incorrect length. We address this with two fixes: (1) numeric scores outside $[-2, 2]$ are clamped to the nearest valid value, preserving ordinal intent; (2) if batch scoring fails after three retries, a per-utterance fallback scores each statement individually. Each quad is scored in two direction-matched batches (the quad structure is defined in Section 4.1), so a topic of 5 quads incurs 10 scoring calls; the fallback triggered on 2 of the 10 calls for Measure V, and those results are included without special treatment.

4 EXPERIMENTAL DESIGN

4.1 Quad Structure

Each experimental unit is a *quad*: four legs run from the same seed, varying condition (control / treatment) and initial suspect direction (−2 or +2). Direction-matching across conditions ensures that observed differences are attributable to the treatment rather than to stance-initial effects. Legs within a quad are scored in matched pairs (legs 0 & 2, legs 1 & 3) by the same Judge call, enforcing a consistent scoring scale within each direction.

4.2 Topics and Dataset

We evaluate on four topics derived from real ballot measures:

- **Housing (Rezoning)**: composite based on California rezoning ballot measures.
- **Arts (Funding)**: composite based on municipal arts levy ballot measures.
- **Transit (Fare-free)**: composite based on fare-free transit ballot measures including Kansas City (2020).
- **Measure V (Village Farms, Davis)**: verbatim arguments from the Yolo County ACE Department for the City of Davis June 2, 2026 Special Municipal Election. This is our real-data condition.

The first three topics use synthesized argument banks modeled after real measures; Measure V uses verbatim supporter and opponent arguments as published by the election authority. All four topics share the same YAML schema and are processed identically by the pipeline.

4.3 Metrics

The Judge scores every suspect utterance, not just the final one. In a 6-turn conversation this yields a trajectory of 7 scores: $s_0, s_1, \ldots, s_6$, where s_0 is the Judge's score of the opening utterance (before the Interrogator has spoken) and s_1–s_6 follow each subsequent suspect reply. Note that s_0 is *measured*, not assigned: it is the Judge's reading of what the suspect actually says first, which need not equal the assigned pole d_0. We compute three metrics over this trajectory:

Magnitude. $|s_6 - s_0|$. The absolute net shift from the suspect's opening stance to their final stance, regardless of direction. The scale runs from 0 to 4; a magnitude of 2.10 means the suspect moved roughly two full points on average.

Directional Accuracy. Fraction of legs where $\text{sign}(s_6 - s_0)$ matches the Interrogator's target direction, i.e., the suspect moved *toward* the Interrogator's position. At 0% the suspect never moved toward the Interrogator (it held its ground or moved away); at 100% it always did; 50% corresponds to direction-agnostic drift. Because our claim is comparative (treatment against its matched control, rather than against an absolute threshold), this metric is robust to the Judge's scale drift: it depends only on the sign of each shift, not its magnitude. This is our primary metric.

Consistency. Variance of $s_0, s_1, \ldots, s_6$ within a single trial. Lower variance means the suspect's stance evolved smoothly or held steady; higher variance indicates erratic flip-flopping between turns. Two trials can share the same final magnitude and direction yet differ sharply in consistency: a trajectory of $[-1, -1, -2]$ (smooth, low variance) is qualitatively different from $[-1, +1, -1]$ (oscillating, high variance) even if both end at −1. Per-turn scoring makes this distinction measurable.

4.4 Scale

We run 5 quads per topic × 4 legs per quad × 6 turns per leg = 120 dialogue turns per topic, 480 turns total. Each topic produces 20 legs (10 control, 10 treatment). The pooled analysis uses all 80 legs (40 per condition).

Reproducibility. All four model calls per turn (suspect, interrogator selector, interrogator generator, judge) use `llama3.1:8b` through a local Ollama server at its default decoding settings (temperature 0.8), with a fixed integer seed supplied on every call. The five quads of a topic use seeds 42–46; within a quad all four legs share the quad's seed, and the same five seeds recur across the four topics. Given the seed, runs are deterministic.

5 RESULTS

5.1 Per-Topic Results

Table 1 reports per-topic metrics. Figure 1 visualizes directional accuracy by condition and topic.

Table 1: Per-topic metrics by condition. Dir. Acc. = directional accuracy (% of legs in which the suspect moved toward the Interrogator). Cons. = consistency (within-trial stance variance; lower = smoother trajectory).

Topic	Condition	Mag.	Dir. Acc.	Cons.
Housing	Control	2.30	40%	1.83
	Treatment	1.60	**80%**	1.09
Arts	Control	1.50	60%	1.30
	Treatment	1.80	60%	1.08
Transit	Control	1.20	10%	1.30
	Treatment	2.10	**90%**	1.27
Measure V	Control	2.00	30%	1.65
	Treatment	1.60	**70%**	1.16
Pooled	Control	1.75	35%	1.52
	Treatment	1.77	**75%**	1.15

Directional accuracy. The headline result is the *pooled* gap: treatment legs move toward the Interrogator in 30 of 40 cases (75%) versus 14 of 40 (35%) for control. A two-proportion test on these counts gives $z = 3.6$, $p < 0.001$, so the gap is very unlikely to be sampling noise. We lead with the pooled comparison deliberately: per-topic differences rest on only $n = 10$ legs per condition, whose 95% confidence intervals span roughly ±30 percentage points, and we therefore treat individual topic rankings as indicative rather than reliable. With that caveat, treatment exceeds control on 3 of 4 topics (Housing, Transit, Measure V) and ties on Arts (60% vs. 60%). The pooled test does carry one qualification: the five seeds recur across topics and the two conditions are matched within a quad, so the legs are not fully independent; a larger fully-crossed run would tighten this estimate.

Transit shows the largest nominal gap (control 10%, treatment 90%), but at $n = 10$ this single-topic number is the least certain in the paper and we do not build any claim on it alone. It does, however, surface a pattern visible across topics: control accuracy is frequently *below* 50%. We examine that in Section 6.2.

Measure V (real data) mirrors the Housing result exactly: +40pp directional accuracy improvement. Despite having a much longer, more complex proposition and 17+ arguments per side (vs. 5 in the synthetic topics), the treatment pipeline produces the same magnitude of improvement. This confirms that the technique recommender generalizes to real-world argument corpora.

Arts is the only topic where treatment does not improve over control. We attribute this to the nature of arts-funding arguments, which tend toward values-based appeals (equity, culture, community) rather than factual claims. The techniques retrieved here lean toward the Reid and Scharff frameworks, which target factual-claim disputes; why the corpus's values-applicable debate cards did not close the gap is examined in the Discussion.

5.2 Magnitude and Consistency

Magnitude effects are mixed (Figure 2): Transit shows a strong positive effect (+0.90), while Housing (−0.70) and Measure V (−0.40) show negative effects. We caution against over-interpreting magnitude results: the Judge LLM (llama3.1:8b) exhibits scale drift on complex or politically charged propositions, assigning systematically higher or lower absolute scores regardless of the actual stance. Directional accuracy, which depends only on the sign of the final shift, is substantially less sensitive to this artifact and is therefore our primary metric.

Consistency (within-trial stance variance) is uniformly lower in treatment legs (≈ 1.15) than control (≈ 1.52). Lower variance means the suspect's stance shifted smoothly rather than flip-flopping between turns, consistent with genuine directional influence. The control interrogator, lacking technique guidance, appears to provoke resistance on some turns and concession on others, producing oscillation; the RAG-guided interrogator produces more monotonic trajectories (Figure 3).

5.3 Figures

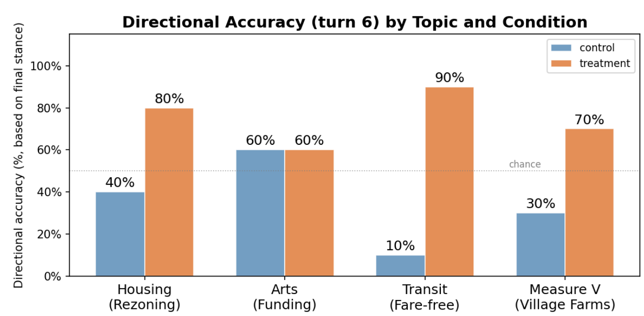

Figure 1: Directional accuracy (%) by topic and condition. The dashed line at 50% marks direction-agnostic drift. Treatment exceeds control on 3 of 4 topics; Arts is the exception (60% = 60%).

6 DISCUSSION

6.1 Why Does Technique Recommendation Help?

The treatment Interrogator has access to two things the control lacks: (1) a phase-appropriate technique that shapes *how* to engage (e.g., Scharff elicitation in the middle phase, which avoids direct contradiction and instead frames counter-arguments as the Suspect's own conclusions), and (2) retrieved topic arguments that give the Interrogator factually grounded claims rather than relying

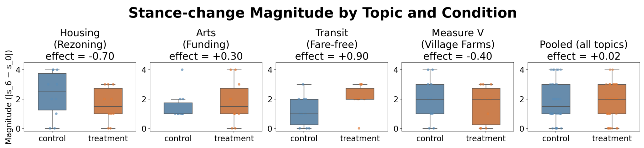

Figure 2: Stance-change magnitude (boxplots with individual points) by topic and condition. The pooled panel (rightmost) shows overlapping distributions with similar means. Magnitude is a secondary metric due to judge scale drift on complex propositions; directional accuracy (Figure 1) is more reliable.

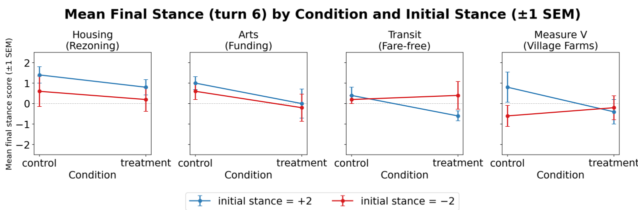

Figure 3: Mean final stance (turn 6) by condition and initial suspect stance (±1 SEM); line colour encodes initial stance (+2 = blue, −2 = red). Treatment generally moves suspects toward the Interrogator's position, but appears less effective than control for initial stance −2 on Housing and Arts. Turn-0 baseline scores are unavailable in the current results. See Figure 1 for directional accuracy.

on the base model's parametric knowledge. The combination produces more coherent, strategically timed responses, which in turn produces more consistent Suspect movement. We caution that this account is a *hypothesis*: because the pipeline does not log which technique the selector chose each turn (Section 6.4), we cannot verify that the generator actually followed the selected technique. A manual audit of transcripts (checking technique adherence on a sample of turns) would be needed to establish the mechanism, and we leave it to future work.

6.2 Why Is the Control Below Chance?

A striking feature of Table 1 is that pooled control accuracy is 35%, *below* the 50% direction-agnostic line, and as low as 10% on Transit. The unguided interrogator is therefore, on average, associated with the suspect moving *away* from its target. We do not have a logged mechanism and offer three non-exclusive candidate explanations. (i) A *backfire* effect: unguided, ungrounded argumentation may read as pushy or weak and entrench the suspect, a well-documented dynamic in human persuasion. (ii) *Bounded-movement geometry*: suspects start at a stance pole ($d_0 = \pm 2$), so once the Judge scores the opening utterance near that pole there is little room to move further toward the target and ample room to move away, biasing the sign of $s_6 - s_0$ against the target under noise. (iii) *Self-revelation*: against a weak interrogator the suspect may simply elaborate its initial position, which the Judge reads as movement away. Because the size of our headline gap depends on control being this low, distinguishing these explanations, ideally with a neutral-start condition ($d_0 = 0$) that removes the pole geometry, is important future work.

6.3 The Arts Exception

The lack of improvement on the Arts topic is instructive. Arts-funding debates center on values ("arts enrich communities", "public money should fund public goods"), which are less amenable to the factual-claim structure that the Reid and Scharff cards target. Our corpus is not, however, devoid of values-applicable tools: the six structured-debate cards (steelmanning, Socratic questioning) apply directly to value disagreements, as does the Motivational Interviewing card. The null result may therefore reflect not a missing capability but a *selection* failure, with the phase-filtered selector not surfacing those cards (or their being insufficient on their own). Distinguishing "wrong tools available" from "right tools not selected" requires the per-turn technique logging we currently lack (Section 6.4), and is an interesting question in its own right.

6.4 Limitations

LLM-vs.-LLM evaluation and judge circularity. All three roles use the same model family (llama3.1:8b). Estornell and Liu [2] show that shared training distributions introduce correlated biases, but in our setting the concern is sharper than bias: the same model distribution generates the suspect's utterances, produces the interrogator's moves, *and* grades the outcome, so the primary metric is partly circular. A held-out Judge from a different family (e.g., qwen2.5:7b) is therefore not merely a refinement but a precondition for fully trusting the absolute numbers. We foreground directional accuracy precisely because it is the measure least sensitive to the Judge's absolute calibration, but it does not escape the circularity entirely.

Judge scale drift. On complex propositions (especially Measure V and Housing), the Judge occasionally scores utterances outside $[-2, 2]$ or returns arrays of incorrect length. We apply clamping and a per-utterance fallback, but the underlying instability inflates variance in the magnitude metric. Directional accuracy is substantially more robust.

Small *N* and independence. The pooled comparison (40 legs per condition) is statistically significant ($z = 3.6$, $p < 0.001$), but per-topic comparisons ($n = 10$) are not powered for significance testing, so individual topic rankings are indicative only. The five quads within a topic use distinct seeds (42–46), yet the same five seeds recur across topics and the two conditions are matched within a quad, so treating the 80 legs as fully independent is an approximation; a larger, fully-crossed run is needed to tighten the estimates.

Technique vs. argument confound. The treatment condition adds two things the control lacks: technique guidance (Channel 1) and retrieved topic arguments (Channel 2). Our design therefore measures their combined effect and cannot isolate the contribution of technique selection on its own. The natural ablation, an arguments-only condition that supplies Channel 2 without Channel 1, would separate the two; we leave it for future work.

Technique logging gap. The current pipeline does not write the selected technique back to the transcript, preventing post-hoc analysis of which techniques were most effective or most frequently selected. This limits our ability to close the recommendation loop and identify underperforming technique cards.

Single-topic real dataset. Measure V is one real topic. A fuller evaluation would use a set of real ballot measures (or social media arguments) across a wider domain range.

Benign topics vs. extremist framing. Our motivation invokes online extremism and entrenched belief, but we evaluate on low-affect civic ballot measures chosen for neutrality and reproducibility. Genuinely extremist or identity-protective stances may resist persuasion through mechanisms (motivated reasoning, identity threat) that are weak or absent here. Our results are therefore best read as evidence about controlled stance change: a proxy for, not a direct test of, moderating entrenched or extremist views.

6.5 Ethical Considerations

This system is designed as a research tool for studying LLM persuasion dynamics, not for deployment against real humans without consent. Techniques drawn from the Reid framework in particular have been criticized for producing false confessions in real interrogation contexts [7]; in our setting they are applied to LLM-simulated suspects in a controlled evaluation. Deployment in any real moderation context would require independent safety review, consent mechanisms, and domain-specific calibration. More broadly, the interrogation framing we borrow from, while technically precise for our technique corpus, can obscure the underlying intent: to create the conditions for genuine reconsideration. A responsible deployment would treat individuals with entrenched views not as targets to be argued into submission, but as people who may have unmet social, psychological, or informational needs. A well-designed system should be capable of shifting from structured dialogue toward supportive, motivational exchange when such needs emerge.

7 CONCLUSION

We have shown that a RAG-powered technique recommender improves LLM interrogator performance on a stance-change task across four ballot-measure topics, including one real-world dataset (Davis Measure V). The treatment condition achieves 75% directional accuracy versus 35% for the unguided control, a 40-percentage-point improvement ($p < 0.001$, pooled); the largest single-topic gap is Transit (10% → 90%), though per-topic samples are small. Technique recommendation also improves stance consistency, suggesting coherent directional influence rather than noise-driven oscillation. The real-data condition (Measure V) replicates the improvement despite substantially richer and more complex argument structure, supporting the generalizability of the approach.

The primary limitation is the LLM-vs.-LLM evaluation setting, which introduces shared-bias confounds. Future work should add an arguments-only condition to isolate the technique channel from the argument channel, evaluate with a held-out Judge model, instrument the system to log selected techniques per turn, and expand the real-data condition to a broader set of genuine public-debate topics.

A broader extension would connect this system to network-scale intervention. Research on online ideological communities shows that influence spreads non-uniformly: a small number of high-centrality actors account for a disproportionate share of content propagation [9], and dynamic graph-learning tools [10, 11] can model how such influence evolves over time. A natural extension would pair network analysis with our dialogue recommender, first identifying likely-responsive high-centrality individuals, then initiating technique-guided conversations with them. We must, however, flag an unresolved tension with the consent principle in our Ethical Considerations: deploying technique-guided persuasion against non-consenting, network-selected targets would be manipulation at scale, which we do not endorse. We frame this direction as legitimate only under transparency and opt-in conditions (for example, users who have explicitly requested a debate partner, or platform tools that disclose the system's nature and purpose). Resolving this tension is a prerequisite to any real deployment, not an afterthought.

8 CONTRIBUTION STATEMENT

Each member wrote their own contribution statement.

Marcin Wróblewski: Initiated the project and its core research idea (using real-world interrogation techniques to moderate extremist stances). Proposed and developed the overall three-role multi-agent architecture (Suspect, Interrogator, Judge), the quad-based experimental design, the real-data generalization test (the Davis Measure V condition) and social network intervention application. Led the team and organized the collaborative working meetings, and maintained the project's GitHub infrastructure. Finalized the slides for the final presentation. Composed proposal document. Contributed to writing and revising the paper, and refining the figures and plots.

Haoyu Yan: Implemented the entire codebase realizing the team's designs: the LLM client (Ollama HTTP wrapper); the three roles, including the Interrogator's control branch and two-stage RAG treatment pipeline; the two-channel RAG recommender (18 technique cards, topic cards, ChromaDB index, and phase-aware retrievers); the quad experiment runner; the metrics; the analysis scripts and figures; and the Measure V data integration. Authored the full paper draft in ACM format (iterative revisions, citation verification, significance analysis, and limitations) and drafted the presentation slides. Contributed to revising the paper.

Bill Steeven Koumba Moussadji Lu: Proposed the evaluation methodology: the magnitude, direction, and consistency metrics and the worked-example Judge scoring scheme. Led the interpretation of the experimental results during the working meetings. Gave detailed editorial review of the paper drafts, with stylistic and structural suggestions that were incorporated into the final version, and reviewed and refined the presentation slides. Contributed to revising the paper.

Sanjay Manivasagam: Built the project's cross-OS portability layer: cross-platform Ollama discovery (scripts/ollama_runtime.py), the UTF-8 console fix for Windows, the one-command setup.py bootstrap, the CI matrix (Ubuntu/macOS/Windows × Python 3.11–3.12), and the portability audit (docs/PORTABILITY.md), enabling reproducible runs on all three operating systems. Authored the project README and contributed early architectural sourcing for the two-agent dialogue design. Conducted a methodological peer review of the paper (judge-independence, the techniques-vs-arguments confound, significance testing, and ablation gaps). Owns the Experimental Design and Evaluation Metrics presentation sections.

9 MEETING PARTICIPATION

The majority of the project's progress was made collaboratively during the team's working meetings. We note attendance below as one objective reference point.

Member	Meetings attended
Marcin Wróblewski	16
Haoyu Yan	12
Bill Steeven Koumba Moussadji Lu	9
Sanjay Manivasagam	5

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